

First Independent Verification of the Boundary Anchoring Method

——Using the Involution Case as the Test Object

He Guorui

Independent Researcher

Corresponding Author: He Guorui

Email: NooEcology@outlook.com

ORCID: 0009-0006-2947-0032

Theoretical Affiliation

This paper belongs to the research program of *Noo-Ecology* and *Civilization Dynamics*.

Methodological Foundation: The Boundary Anchoring Method, first articulated in the Methodological Appendix of *Noo-Ecology* (2026), and subsequently incorporated into *Civilization Dynamics* (2026). It will be visible with the next version update.

This paper constitutes the first independent verification of the method.

Methodological Declaration

This paper employs the Boundary Anchoring Method as its core verification tool. The verification logic follows the pathway of "left-end anchoring → deviation ratio calculation → comparison against independent scholarly benchmarks." The pre-specified verification failure criteria are publicly stated in the Introduction to prevent circular reasoning.

Version Information

First draft completed in April 2026. Subsequent revisions will incorporate feedback from verification results.

Public Access to the Theoretical System

The full texts of *Noo-Ecology* and *Civilization Dynamics*, along with revised versions of this paper, are available via:

GitHub: <https://github.com/wenhe1994/wenhe001>

Zenodo: <https://doi.org/10.5281/zenodo.19680835> (Noo-Ecology)

Zenodo: <https://doi.org/10.5281/zenodo.19680783> (Civilization Dynamics)

Abstract

Problem Background

Grand social theories have long faced a methodological dilemma: their core variables——such as class consciousness, social solidarity, the diffusion of power, and meta-cognitive level——are highly self-consistent within the theoretical framework, yet are extremely difficult to measure and operationalize directly. The Boundary Anchoring Method was proposed to address this very difficulty by offering an alternative strategy: instead of directly measuring the theoretical variables that define a healthy state, one measures the system's distance from its observable decline limit. The method has been fully articulated in *Noo-Ecology*, but it has not yet undergone independent verification. This paper is its first independent test.

Verification Strategy

Agricultural involution was selected as the verification case for three reasons. First, the decline limit of involution——the diminishing marginal return of labor approaching zero——has already been clearly demarcated with observable indicators by Huang Zongzhi and other researchers. Second, the divergent judgments issued by different scholars on the same system——Huang Zongzhi's "involution thesis," Kenneth Pomeranz's "rejection thesis," and Xu Hao's "extensive farming thesis"——constitute an independently citable set of scholarly benchmarks. Third, three systems——the Jiangnan region, the North China Plain, and Tokugawa Kinai——provide a natural gradient from "involution" to "non-involution," making it possible to test the method's discriminative capacity.

Verification Execution and Results

Taking the daily labor remuneration per unit of rice cultivation in Jiangnan, as given by Huang Zongzhi, as the left-end anchor ($V_{\text{anchor}} \approx 0.287$ shi/labor-day), and using the ratio method $D_i = \frac{V_i}{V_{\text{anchor}}}$ as the computational tool, we calculated the relative labor remuneration ratio (hereinafter the **deviation ratio**) for three systems. The results are as follows: (1) the deviation ratio for Jiangnan cotton-textile sideline production is 19.5%, pressing close to the involution limit; (2) the deviation ratio for poor-peasant family farms in North China is 42% (as directly given in Huang Zongzhi's original work), exhibiting a clear distance from the limit; (3) Tokugawa Kinai agriculture shows no diminishing marginal returns to labor, its deviation ratio moving away from the limit. The rank-ordering of these three results is broadly consistent with mainstream judgments in agricultural history.

Conclusion and Significance

The Boundary Anchoring Method has passed its first independent verification. Using a single indicator——the ratio of daily labor remuneration per unit——and only three data points, it successfully produced a differentiated positioning of the degree of involution across three systems. The passage of this verification provides methodological legitimacy for the method's subsequent cross-domain applications——including the extension to the multi-indicator system of Complex Metabolic Failure. This paper constitutes the first logical building block on the path of the Boundary Anchoring Method from a methodological concept to an operational tool.

Keywords: Boundary Anchoring Method; involution; methodological verification; deviation ratio;

Huang Zongzhi; agricultural economic history

1 Introduction: The Operationalization Barrier in Grand Theory and the Task of This Verification

1.1 The Problem of Non-Operationalizability

Grand social theories have long faced an uncomfortable reality: the most profound theories are often built upon the least measurable variables.

Marx's "class consciousness"——how does one measure the degree to which a group has become aware of its own historical position?

Weber's "charisma"——how does one calibrate the intensity of an authority believed to possess extraordinary qualities?

Durkheim's "social solidarity"——how does one empirically distinguish the tipping point between mechanical and organic solidarity?

These concepts are entirely self-consistent within their respective theoretical frameworks. Each supports a grand framework for explaining human collective behavior. Yet the moment one asks, "What is its numerical value? What is its unit? How can it be compared across cases on an interval scale?"——the theorist can only fall back on interpretation, case studies, and intuition.

This is not a failing of any particular theorist. It is a structural dilemma. What grand social theories attempt to capture are often emergent properties of complex systems——holistic states generated from countless micro-level interactions that are irreducible to the mere sum of individual actions. Such states can be clearly defined at the conceptual level, yet they lack a direct counterpart at the operational level. One can feel the "diffusion of power" in a society, but one cannot measure it the way one measures temperature. One can argue that a civilization is undergoing a "crisis of meaning," but one cannot produce a numerical indicator comparable to the GDP growth rate.

The same dilemma appears in *Civilization Dynamics*. That theory proposes a phase-transition potential function to describe the dynamical tendency of a civilization to leap from a "collective mind" to a "collaborative cognitive network." The variables of the function——the meta-cognitive level differential, the proportion of awakened individuals, the systemic stability-suppression coefficient——are strictly self-consistent within the theory. Each is supported by the axiomatic system, and the functional relationships among them can withstand logical scrutiny. Yet when asked, "How does one measure the average meta-cognitive level of a society?" the theory finds itself caught in the same operational suspension that afflicts all grand theories.

This is not a defect of theory. The first mission of theory is to understand; measurement comes second. But a theory that cannot be measured cannot be tested. And a theory that cannot be tested remains permanently suspended in the gray zone between a respectable conjecture and an effective tool. *Civilization Dynamics* seeks to cross this zone. It needs a bridge from metaphorical formulas to

operational formulas.

1.2 The Core Strategy of the Boundary Anchoring Method

The Boundary Anchoring Method was proposed precisely in response to this difficulty.

The method was first articulated by the present author in the Methodological Appendix of *Noo-Ecology* (2026) and subsequently incorporated into *Civilization Dynamics* (2026). It is not a technical toy invented from thin air, but the product of sustained reflection on two questions: If the healthy state of a system cannot itself be directly measured, can we indirectly locate health by measuring the *loss* of health? And if the variables of a grand theory cannot be operationalized, can we instead operationalize the *system-level consequences* produced by the absence of those variables?

The answer offered by the Boundary Anchoring Method is this: For systems whose healthy state cannot be directly measured, yet whose decline limit can be demarcated by observable indicators——do not directly measure the health-defining variables themselves; instead, measure the system's distance from its decline limit. The "meta-cognitive level" of a society may be impossible to measure directly; but when it continuously decays, it leaves observable traces at the level of meaning——for instance, a generational decline in the explanatory coverage of the meta-narrative. Measuring this trace is equivalent to measuring the "degree of meta-cognitive deficit." And the degree of deficit, in turn, carries information about the degree of health.

This is not a replacement for grand theory, but a supplement to it. The Boundary Anchoring Method does not ask, "What is the absolute value of the meta-cognitive level?" It asks, "How far is this society from a meta-cognitive collapse?" The former is a metaphysical question that cannot be answered with precision; the latter is, in principle, an empirical question that can be tested with data.

The full articulation of the method——including its conditions of applicability, its ontological presuppositions, its five-step operational procedure, its normative self-declaration, its relationship to existing methodological traditions, an application example, its failure boundaries, and its mechanism for revision——has been given in the Methodological Appendix of *Noo-Ecology*. We shall not recapitulate it here. The task of the present paper is to move the method from concept to test.

1.3 The Task of This Paper: First Independent Verification

A methodological tool, before being formally deployed, must first demonstrate that it is not ineffective.

This is not an unreasonable demand to place upon a new method. On the contrary, it is a form of intellectual honesty: a method must actively expose the conditions under which it might fail, and then undergo testing under those very conditions. If it passes, it earns the qualification to be extended to more complex domains. If it fails, all subsequent applications are, for the time being, untenable.

This paper is the first independent verification of the Boundary Anchoring Method. Agricultural involution was selected as the verification case not because it is directly relevant to *Civilization Dynamics*, but because it provides one of the very few domains that satisfy the conditions for verification: First, the decline limit of involution has already been clearly demarcated by an independent researcher.

Huang Zongzhi defined the core mechanism of involution as "total output expanding at the cost of diminishing marginal returns per unit of labor per day," and provided a set of quantitative indicators that can be directly extracted——rice cultivation requires 11.5 labor-days per mu and yields 3.0--4.6 *shi* per mu; the labor input of cotton-textile sideline production is 18 times that of rice, yet its total return is only 3--4 times that of rice, meaning that the daily labor remuneration is drastically reduced. This set of data was not constructed by the present paper; it is the very empirical evidence Huang Zongzhi himself used to argue for the existence of involution. Within the framework of the Boundary Anchoring Method, they are reinterpreted as the basis for calculating the left-end anchor value. The authority for calibrating the anchor does not rest with the author of this paper, but with the independent researcher.

Second, different scholars have issued divergent judgments on the same systems, thereby constituting a diverse set of independently citable scholarly benchmarks. Huang Zongzhi himself identified Jiangnan as a case of classic involution, and North China as having a milder degree of involution; Xu Hao, by contrast, argued that North China is not a case of involution at all, but rather one of extensive farming; Kenneth Pomeranz went so far as to completely reject the existence of involution in eighteenth-century Jiangnan, contending that its economic level was on par with that of England. These disagreements are not "noise" to be smoothed over——they are precisely what provide reference coordinates for the method's positioning from multiple angles. If a set of computational results can establish reasonable relationships with these mutually conflicting judgments——not by submitting to the authority of any one school, but by using a single yardstick to mark out which segment of the spectrum each school's gaze has captured——then the explanatory power of the anchoring method is far more compelling than if it were merely consistent with a single judgment.

Third, three systems——Jiangnan, North China, and Tokugawa Kinai——provide a natural gradient. Jiangnan is widely recognized as a classic case of involution; the degree of involution in North China is contested but generally agreed to be milder than Jiangnan, or to belong to a different type; and Kinai has been confirmed by independent scholarship as a non-involution case of intensive development. This gradient allows the method to be directly tested on its capacity to distinguish different degrees of systemic state.

The verification logic of this paper is as follows. Step one: take the daily labor remuneration per unit of rice cultivation in Jiangnan, as provided by Huang Zongzhi, as the left-end anchor (V_{anchor}). Step two: use the ratio-method formula $D_i = V_i/V_{\text{anchor}}$ to calculate the deviation ratios of the three target systems——the lower the deviation ratio, the closer the system is to the involution limit. Step three: compare the rank-ordering of the computational results against mainstream judgments in agricultural history. If the rank-ordering is systematically consistent with the independent scholarly benchmarks, the verification is passed. If, conversely, the rank-ordering is in systematic conflict with the independent benchmarks and the conflict cannot be eliminated by correcting the anchor value, the verification has failed.

A necessary clarification: The core objective of this verification is to test the **ordinal capacity** of the Boundary Anchoring Method——that is, whether the method, given a pre-specified anchor, can correctly differentiate the degree to which different systems are approaching the decline limit. What is being tested is the operational logic of the method, not the absolute truth of the anchor value. The anchor

is taken from Huang Zongzhi because Huang Zongzhi's original data are currently the only available historical benchmark in the academic literature that includes quantifiable indicators. This does not mean that the method is intrinsically dependent on Huang Zongzhi's theoretical framework. Should future researchers recalibrate the anchor from entirely independent primary sources, the Boundary Anchoring Method can directly deploy the new anchor without requiring any modification to the core operations of the method. This feature of "anchor replaceability" is precisely what distinguishes the Boundary Anchoring Method from fixed theoretical models.

1.4 Pre-Specified Criteria for Verification Failure

To prevent verification from degenerating into circular reasoning——where the very results produced by the method are used to prove the method effective——this paper must publicly state the criteria for verification failure before the verification begins.

This verification shall be judged as "not passed" if, and only if, the following two conditions are simultaneously satisfied:

First, the rank-ordering of the deviation ratios is in conflict with the systematic direction of the independent scholarly benchmarks. Specifically: (a) the deviation ratio for North China turns out to be *lower* than that for Jiangnan (meaning North China is closer to the involution limit than Jiangnan), while mainstream judgment in agricultural history consistently holds that Jiangnan's degree of involution is deeper; or (b) Kinai likewise exhibits significant diminishing marginal returns to labor, while the independent literature unanimously identifies Kinai as a case of intensive development without involution. If either (a) or (b) occurs, an ordinal conflict is established.

Second, the conflict cannot be eliminated by correcting the anchor value. If the conflict arises merely from a technical deviation in the calibration of the anchor——for example, if the choice of the median value for rice yield per mu affects V_{anchor} , and the conflict disappears once the chosen value is adjusted——then the verification is not judged as failed on this account. The verification failure criterion targets the core mechanism of the method, not the auxiliary operation of anchor calibration, which can itself be independently tested.

If both conditions are simultaneously satisfied, the Boundary Anchoring Method has not passed this verification. In that case, it will be necessary to trace the source of the failure: Is it a problem with the design of the ratio-method formula itself? Is it that the indicator of daily labor remuneration per unit is insufficient to capture the essence of involution? Or is there a fundamental defect in the demarcation of the decline limit's anchor? This tracing process will itself constitute the starting point for a revision of the method. If, conversely, the computational results are consistent with the independent scholarly benchmarks, the method passes the verification and acquires the initial legitimacy required for extension to more complex domains.

This setting of "pre-specified criteria" is a manifestation of the intellectual honesty intrinsic to the Boundary Anchoring Method. A method that cannot even be conceived of as possibly wrong is either a tautological logical proposition or an untestable piece of metaphysical speculation. The Boundary Anchoring Method is neither. It can be tested, and it is willing to expose itself to the risk of failure.

2 Verification Method

2.1 The Ratio-Method Formula

This paper adopts the ratio method to calculate deviation ratios. The formula is:

$$D_i = \frac{V_i}{V_{\text{anchor}}}$$

where:

- D_i : the deviation ratio of target system i . The closer D_i is to 1, the closer the system is to the involution limit; the smaller D_i is, the more severe the discount on daily labor remuneration relative to the non-involution benchmark, and thus the deeper the degree of involution. Hereafter, D_i is referred to uniformly as the deviation ratio.
- V_i : the daily labor remuneration per unit of the target system.
- V_{anchor} : the left-end anchor value——taken as the daily labor remuneration per unit at the involution limit state.

The advantage of the ratio method lies in its dimensionless nature. Remuneration values from different systems, different periods, and different currencies cannot be directly compared by subtraction, but ratios are immune to differences in units. The judgment that a system's daily labor remuneration per unit is 42% of its non-involution benchmark does not depend on whether that remuneration is denominated in silver dollars, *shi* of rice, or copper coins.

This paper selects daily labor remuneration per unit as the sole indicator for verification, not because it is the only manifestation of involution, but because it is the most direct, the most central, and the least controversial indicator——Huang Zongzhi himself defined involution precisely in terms of diminishing marginal returns to labor. Weighting multiple indicators together is a step for the expansion phase, after the method has been shown to work in an initial test; it is not appropriate to bundle such weighting together with the initial verification.

It is necessary to explain that while the absolute values of Jiangnan rice cultivation and North China managerial farming cannot be directly compared (they are in different units), they play the same role within their respective systems——that of a reference benchmark for the rational allocation of labor under market conditions. Huang Zongzhi used rice cultivation as the baseline form of labor throughout his book; the implicit logic is that the daily labor remuneration per unit of rice cultivation represents the level of "normal market returns" in Jiangnan agriculture, against which the remuneration of sideline activities such as cotton spinning and sericulture is compared. Similarly, in North China, managerial farms hire labor at market wages and cease to add further labor once the marginal return drops below the wage level——the daily labor remuneration per unit of a managerial farm represents the "market wage" level in North China agriculture. The economic function of each within its own region is isomorphic: each is the steady state in which "labor is freely mobile and remuneration is determined by the market." This isomorphism makes it logically possible to place the deviation ratios obtained from the internal comparisons of the two systems side by side——not as a conversion between *shi* and silver dollars, but as a comparison of "how far each is from its own normal line."

2.2 Rationale for Case Selection

This paper selects agricultural involution as the verification case, not because involution is directly relevant to the ultimate domain of application of the Boundary Anchoring Method (*Civilization Dynamics*), but because involution provides one of the very few verification fields that simultaneously satisfy the following three conditions.

Condition One: The decline limit has already been clearly demarcated by an independent researcher. In *The Peasant Family and Rural Development in the Yangzi Delta*, Huang Zongzhi defined the core mechanism of involution as "total output expanding at the cost of diminishing marginal returns per unit of labor per day," and provided a set of quantitative data that can be directly extracted——rice cultivation requires 11.5 labor-days per mu and yields 3.0--4.6 *shi* per mu; the labor input of cotton-textile sideline production is 18 times that of rice, yet its total return is only 3--4 times that of rice. This set of data was not constructed by the present paper; it is the very empirical evidence Huang Zongzhi himself used to argue for the existence of involution. Within the framework of the Boundary Anchoring Method, they are reinterpreted as the basis for calculating the left-end anchor value. The authority for calibrating the anchor does not rest with the author of this paper, but with the independent researcher.

Condition Two: Different scholars have issued mutually divergent judgments on the same systems, thereby constituting an independently citable set of scholarly benchmarks. Huang Zongzhi identified eighteenth-century Jiangnan as a case of classic involution, with North China having a milder degree of involution but still exhibiting the same type; Xu Hao, by contrast, argued that North China is not a case of involution at all, but rather one of extensive farming——low input, low output, but stable marginal returns; Kenneth Pomeranz went so far as to completely reject the existence of involution in eighteenth-century Jiangnan, contending that its economic level was on par with that of England. These disagreements are not "noise" to be reconciled——they are precisely what provide reference coordinates for the anchoring method's positioning from multiple angles. If a set of computational results can establish reasonable relationships with these mutually conflicting judgments——not by submitting to the authority of any one school, but by using a single yardstick to mark out which segment of the spectrum each school's gaze has captured——then the explanatory power of the anchoring method is far more compelling than if it were merely consistent with a single judgment.

Condition Three: The existence of natural gradient cases. Jiangnan (classic involution), North China (degree of involution contested but generally agreed to be milder than Jiangnan, or to belong to a different type), and Tokugawa Kinai (confirmed by independent scholarship as a non-involution case of intensive development) form three tiers, making it possible to directly test the method's capacity to distinguish different degrees of systemic state.

2.3 Sources of the Independent Scholarly Benchmarks

The Boundary Anchoring Method requires that verification results be compared against independent scholarly judgments, not against results produced by the method itself in a circular fashion. This paper invokes the following independent sources as comparative benchmarks:

System Scholarly Judgment Source

Jiangnan Classic involution; severe diminishing marginal returns to labor Huang Zongzhi, *Yangzi*

Delta (1992);

cotton- Huang Zongzhi, "Revisiting Involution"

textile (2021)

region

North China Semi-involution/extensive farming. Huang Zongzhi considers the degree of involution milder than Jiangnan; Huang Zongzhi, *North China* (2023);

poor- Xu Hao explicitly identifies it as extensive farming, not involution Xu Hao (2000)

peasant

farms

Tokugawa Intensive development; increasing daily labor remuneration per unit; no involution Smith (1959); Hanley & Yamamura (1977)

Kinai

Kenneth Pomeranz's (2000) fundamental rejection of the existence of involution in Jiangnan does not directly serve as a benchmark for any specific system in this verification, but it does constitute an ontological challenge to the anchor itself——what he questions is not "how far Jiangnan is from the involution limit on the spectrum," but "whether the involution limit can be described in the way Huang Zongzhi described it at all." The Boundary Anchoring Method does not enter into this ontological dispute. It accepts Huang Zongzhi's anchor as an operational premise (because this anchor can be independently tested and already has a mature body of scholarly discussion surrounding it), and on that basis performs the positioning operation. If the anchor itself is overturned by future research, the anchoring method need only recalibrate the anchor; the method's core mechanism remains undamaged——this is precisely the advantage that distinguishes the anchoring method from fixed theoretical models.

Given that the units of measurement are not fully unified across the three systems——Jiangnan uses *shi*/labor-day, North China uses *yuan*/labor-day, and for Kinai we have only trend data on productivity without a single numerical value for daily remuneration——the verification in this paper is executed on two levels. Between Jiangnan and North China, comparison is made through their respective "internal ratios" (i.e., deviation ratios): the internal ratio for Jiangnan is the ratio of cotton-textile sideline remuneration to the rice-cultivation benchmark; the internal ratio for North China is the ratio of poor-peasant farm remuneration to the managerial-farm benchmark. The two internal ratios are comparable under the same logic——each is "involution remuneration / non-involution remuneration." For Kinai, because the data take the form of trend statements rather than cross-sectional ratios, ordinal positioning is used: $D_{\text{Kinai}} \gg D_{\text{North China}} > D_{\text{Jiangnan}}$.

2.4 Basic Logic of the Verification

To summarize the above, the execution logic of this verification is as follows. Step one: we take the daily labor remuneration per unit of rice cultivation in Jiangnan, as provided by Huang Zongzhi, as the left-end anchor (V_{anchor}), and simultaneously calculate the internal cotton-textile deviation ratio for Jiangnan to confirm the degree of involution at the anchor region. Step two: we calculate the internal deviation ratio of poor-peasant farms in North China, as target system one. Step three: we rely on the qualitative characterizations of Kinai agriculture in the independent literature to determine the direction and degree of its deviation ratio, as target system two. Step four: we compare the rank-ordering of the deviation ratios of the three systems against the independent scholarly benchmarks.

The conclusion of this logical chain is not "whether involution exists," nor "whose theory of involution is correct." The conclusion is: Can the Boundary Anchoring Method, using only a single indicator and only three data points, produce a differentiated positioning that is consistent with mainstream scholarly judgments? If it can, it has demonstrated its preliminary effectiveness as a positioning tool. If it cannot, it needs to be revised or abandoned.

3 Left-End Anchoring: Calibrating the Involution Limit State in

Jiangnan

3.1 Data Sources

The anchor data are taken from Chapter Five, "Commercialization and Involutionary Growth," of Huang Zongzhi's *The Peasant Family and Rural Development in the Yangzi Delta* (Zhonghua Book Company, 1992 edition). In that chapter, Huang provides three sets of original numerical values that can be used for anchor calibration.

Rice cultivation. It requires 11.5 labor-days per mu, and yields 3.0--4.6 *shi* of paddy rice per mu (1 *shi* $\approx$ 120 *jin*). Rice cultivation serves as the baseline form of labor throughout Huang's book——whenever he compares sideline activities such as cotton spinning and sericulture, he consistently uses the daily labor remuneration per unit of rice cultivation as the reference. Within the logic of the present book, the daily labor remuneration per unit of rice cultivation is defined as the "non-involution benchmark."

Sericulture sideline production. The labor input for mulberry cultivation and silkworm rearing reaches 93 labor-days per mu; by comparison, rice cultivation requires only 11.5 labor-days per mu——a labor-input ratio of 8.1:1. Yet Huang states unequivocally: "The net income per labor-day of sericulture is in fact far lower than that of rice cultivation, unless silk prices reach an unusually favorable level."

Cotton-textile sideline production. The total labor-days invested per mu in cotton-spinning-and-weaving production reaches 18 times that of rice, yet the total income is only 3--4 times that of rice. More critically, Huang adds a supplementary observation on the labor composition of cotton-textile sideline production: the daily income from spinning is sufficient to feed only a child under the age of ten; almost all such labor is performed by the elderly and children, and an adult male would never work for such remuneration. This detail implies that the daily labor remuneration per unit of cotton-textile sideline production is not merely "lower than that of rice," but "lower than the subsistence wage of an adult male laborer."

Huang subsumes the above phenomena under a single definition: "Involution (*guomihua*): total output expanding at the cost of diminishing marginal returns per unit of labor per day." This definition is the logical basis upon which the present paper anchors the left end——the hallmark of the involution limit state is not a decline in total output, but a continuous decline in remuneration per unit of labor, to the point where it can no longer even cover a normal subsistence wage.

3.2 Anchor Value Calculation

Taking 3.3 *shi* as the median value for yield per mu, the daily labor remuneration per unit of rice cultivation is calculated as:

$$V_{\text{anchor}} = \frac{3.3}{11.5} \approx 0.287 \text{ shi/labor-day}$$

This value is the non-involution labor benchmark implicit throughout Huang Zongzhi's book. It was not derived by the present paper; it is the reference frame that Huang himself established——he simply never marked it with the symbol " V_{anchor} ." What the present paper does is to make this implicit benchmark explicit by placing it into the formula as the anchor value.

Taking 3.5 as the median multiple for total return, the daily labor remuneration per unit of cotton-textile sideline production is calculated as:

$$V_{\text{cotton-textile}} = \frac{3.3 \times 3.5}{11.5 \times 18} = \frac{11.55}{207} \approx 0.056 \text{ shi/labor-day}$$

The internal deviation ratio for Jiangnan is therefore:

$$D_{\text{Jiangnan}} = \frac{0.056}{0.287} \approx 19.5\%$$

The daily labor remuneration per unit of cotton-textile sideline production is only about one-fifth of the rice-cultivation benchmark. This means that the remuneration a laborer can earn in one day of rice cultivation requires five days to earn in cotton-textile sideline production. This is the quantitative expression of precisely what Huang Zongzhi described as "diminishing marginal returns per unit of labor per day."

3.3 Qualitative Confirmation of the Anchor

Beyond the quantitative calculation, Huang Zongzhi's qualitative characterizations of sericulture and cotton spinning provide indispensable support for the interpretation of the anchor.

Although the daily labor remuneration per unit of sericulture is far lower than that of rice, sericulture still possesses a degree of market elasticity and a skill premium——in years of favorable silk prices, the remuneration per unit can rise significantly. This means that sericulture is not a pure case of involution; under specific market conditions, it can partially escape the trap of diminishing marginal returns.

Cotton spinning is different. The daily income from spinning is sufficient to feed only children; no adult male would work for such remuneration. This means that the daily labor remuneration per unit of cotton spinning has fallen below the subsistence wage of a family——the only reason it can continue to exist is that the peasant family has no option to "lay off" its auxiliary laborers. The elderly, women, and children have zero opportunity cost; they cannot migrate out to seek other work, and thus must accept any positive remuneration, however low.

This distinction carries significant methodological implications for the Boundary Anchoring Method. The limit anchor should not be placed at a state that still retains market elasticity, still commands a skill premium, or still has the potential to rebound. The limit anchor should be placed at a state that has already fallen below subsistence wages and is sustained only by the "inability of labor to exit." The Jiangnan cotton-textile sector is a classic instance of the latter. The present paper therefore treats the deviation

ratio of the Jiangnan cotton-textile region ($D = 19.5\%$) as the reference point for "approaching the involution limit."

3.4 Methodological Response to the Pomeranz Objection

Kenneth Pomeranz's fundamental challenge to Huang Zongzhi's involution framework in *The Great Divergence* bears directly on the ontological status of the anchor in this paper. If Jiangnan does not in fact exhibit the involution limit that Huang describes——if all of Huang's indicators are empirically distorted——then 0.287 shi/labor-day no longer corresponds to any decline limit. The anchor of this paper would lose the object it purports to calibrate.

Yet the Boundary Anchoring Method would not thereby be invalidated.

The core operation of the Boundary Anchoring Method is this: starting from indicators of a decline limit, calculate the distance of target systems from that limit. The structure of this operation does not depend on the ontological truth of any particular anchor. If Pomeranz is correct——if Huang Zongzhi's anchor is found not to correspond to a real involution limit——the Boundary Anchoring Method need not be overthrown. It need only be recalibrated: scholars can independently calibrate a new decline limit from primary archival materials (for example, using the family subsistence wage as a standard to determine at what level of labor remuneration a family can no longer maintain simple reproduction), and then substitute the new anchor into the ratio-method formula to recompute the deviation ratios.

The anchor changes; the structure of the formula remains the same. The criteria for selecting indicators remain the same. The core mechanism of the method is undamaged.

This marks a fundamental difference from traditional theoretical models that rely on fixed variables. In a traditional model, if the definition of a core variable is overturned, the entire parameter system of the model must be reconstructed. The "anchor-deviation ratio" structure of the Boundary Anchoring Method, by contrast, designs the anchor as a replaceable interface——it is not a fixed component of the theory, but an input port of the theory. Any future researcher who provides a new, independently calibrated decline limit can connect it to the same positioning algorithm and obtain a new rank-ordering of deviation ratios.

The Pomeranz objection, therefore, does not constitute a counterexample to this verification. What it touches is the ontological status of the anchor, not the operational logic of the method. And the replaceability of the anchor is precisely the core feature that the Boundary Anchoring Method was designed from the outset to incorporate——it enables the method both to take existing scholarly consensus as its starting point and to avoid being permanently bound to any particular consensus.

3.5 Chapter Summary

The left-end anchor has been calibrated.

Anchor value: $V_{\text{anchor}} = 0.287$ shi/labor-day, taken from the original numerical values for daily labor remuneration per unit of rice cultivation in Huang Zongzhi's work.

Degree of involution at the anchor region: the deviation ratio for Jiangnan cotton-textile sideline production is 19.5%. What this value signifies is this: in a system where daily labor remuneration per unit is merely one-fifth of the non-involution benchmark, laborers can no longer improve their livelihood in proportion to the additional labor they invest, yet the structural constraints of the family make it impossible for them to stop investing additional labor. This is precisely the involution limit that Huang Zongzhi described.

Next, we shall calibrate the deviation ratio of poor-peasant farms in North China using the same indicator logic. This will be the Boundary Anchoring Method's first cross-system positioning——using the indicator framework represented by the Jiangnan anchor to measure the state of North China. If the deviation ratio of North China differs significantly from that of Jiangnan, and if the direction of the difference is consistent with the independent scholarly benchmarks, the discriminative capacity of the method will have passed its first test.

4 Target System One: North China

4.1 Data Sources

The North China data are taken from Huang Zongzhi's *The Peasant Economy and Social Change in North China* (Guangxi Normal University Press, 2023 edition). Unlike the Jiangnan section, where the core data were concentrated in Chapter Five, the core data for North China are concentrated in Chapter Six, which draws on the detailed survey data of Michang Village (one of the villages covered by the Mantetsu surveys) to systematically compare labor productivity across different types of farms.

In Chapter Nine of the book, "A Comparison of Labor Productivity between Managerial and Family Farms," Huang presents the quantitative centerpiece of the entire work——Table 9.2, "Labor Input and Income per Mu by Stratum in Michang Village (1937)." The table arrays five types of farming households horizontally——managerial farmers, rich peasants, middle peasants, poor peasants, and hired laborers——and three indicators vertically——average labor-days per mu, average total income per labor-day (in *yuan*).

The data are as follows:

Farm Type	Labor-Days per Mu (avg.)	Total Income per Labor-Day (yuan, avg.)
Managerial farm	11.0	1.46
Rich peasant	16.6	1.10
Middle peasant	16.8	0.81
Poor peasant	19.3	0.62

The analysis in this section takes this table as its foundation.

4.2 A Structural Reading of the Data

Table 9.2 has three layers of structure, each of which directly serves the calculation of the deviation ratio.

Layer one: the "rational benchmark" of the managerial farm. The managerial farm operates with hired labor; it hires at market wage rates and ceases to invest additional labor once the marginal return drops below the wage. Its labor-days per mu (11.0 days) are the lowest among the five types, and its total income per labor-day (1.46 *yuan*) is the highest. Within Huang Zongzhi's logic, the managerial farm represents the level of rational labor allocation under market conditions—it has no surplus labor input and no suppressed remuneration. This is entirely isomorphic with the role played by Jiangnan rice cultivation in *The Yangzi Delta*: both serve as the "non-involution benchmark" within their respective regions.

Layer two: the "intermediate states" of rich and middle peasants. Rich peasants (16.6 days/mu, 1.10 *yuan/day*) and middle peasants (16.8 days/mu, 0.81 *yuan/day*) fall between the two extremes. They possess a certain amount of land and capital, yet must also invest more labor than the managerial farm to sustain their income. Their presence demonstrates that North China agriculture is not a simple bipolar structure—with normal managerial farms on one side and limit-involution poor-peasant farms on the other. There is a continuous gradient.

Layer three: the "involution value" of the poor-peasant farm. The poor peasant invests 19.3 labor-days per mu——1.75 times that of the managerial farm——yet the total income per labor-day is only 0.62 *yuan*, a mere 42% of the managerial farm's figure. Driven by subsistence pressure, the poor peasant continuously invests additional labor on limited land——weeding more meticulously, plowing more deeply, intercropping more intensively——but the additional labor yields only a negligible increase in output. "Labor input above the normal range can only bring diminishing marginal returns." Huang Zongzhi delivers this conclusion in the discussion immediately following Table 9.2.

4.3 Deviation Ratio Calculation

Taking the managerial farm as the "rational-operation benchmark" (the North China analogue of V_{anchor}), and the poor-peasant family farm as the involution target value, the deviation ratio is:

$$D_{\text{North China}} = \frac{0.62}{1.46} \approx 42\%$$

This ratio was not computed by the present paper; it is given directly by Huang Zongzhi in the original work. What the present paper does is to reinterpret it, within the framework of the Boundary Anchoring Method, as "the deviation ratio of the North China poor-peasant farm relative to its local non-involution benchmark."

4.4 Side-by-Side Comparison with Jiangnan

The deviation ratios of Jiangnan and North China can now be aligned under the same logic.

The deviation ratio of Jiangnan cotton-textile sideline production is 19.5%——daily labor remuneration per unit is only one-fifth of the Jiangnan rice-cultivation benchmark. The deviation ratio of the North China poor-peasant farm is 42%——daily labor remuneration per unit is somewhat over two-fifths of the North China managerial-farm benchmark. The commonality between the two is immediately apparent: in both cases, the labor remuneration of the family farm is significantly lower than

its respective local benchmark. The difference between the two is equally apparent: the extent of the discount at the Jiangnan involution state (approximately 80%) is markedly more severe than that in North China (approximately 58%).

Placing the two values side by side yields a rank-ordering:

$$D_{\text{Jiangnan}} (19.5\%) < D_{\text{North China}} (42\%)$$

Jiangnan is closer to the involution limit. North China is at a noticeable distance from the limit.

4.5 Comparison against Independent Scholarly Benchmarks

The degree of involution in North China has not commanded the same scholarly consensus as that in Jiangnan. Different scholars have issued divergent judgments on the same system.

Huang Zongzhi's own judgment: involution exists in North China, but its degree is milder than in Jiangnan. His formulation describes Jiangnan as "classic involution" and North China as "a differentiating peasant economy"; the logic of involution holds in both regions, but North China's population density, degree of commercialization, and structure of labor input have prevented involution from developing to the extreme seen in Jiangnan.

Xu Hao's judgment: North China is not a case of involution, but of extensive farming. In his paper "On the Extensive Farming of North China Agriculture in the Qing Dynasty," Xu argues explicitly that the defining feature of North China agriculture is not intensive cultivation or involution, and that "using the 'involution' theory to explain the backwardness of labor productivity in North China does not accord with the historical facts of the Qing period." The problem in North China lies in extensive sowing with meager returns, poor soil, backward technology, and insufficient labor input—it is a different path to low efficiency.

Kenneth Pomeranz's judgment: the involution of Jiangnan does not exist in the first place, and even less so that of North China.

These three judgments constitute an independent benchmark set against which the positioning results of the anchoring method can be tested. The computational results of this paper—a 42% deviation ratio for North China, significantly higher than the 19.5% of Jiangnan—relate to these three judgments as follows:

Consistent with Huang Zongzhi's judgment: North China's 42% > Jiangnan's 19.5%. The degree of involution in North China is indeed milder; the two are not on the same segment of the same diminishing-returns curve. The rank-ordering produced by the anchoring method fully matches Huang's own qualitative characterization.

Compatible with Xu Hao's judgment: 42% is not 19.5%. The daily labor remuneration per unit of the North China poor peasant is lower than that of the local managerial farm, but the extent of the discount is far milder than that of the Jiangnan cotton-textile sector. The value 42% does not in itself assert that North China "is a case of involution"—it only asserts that the remuneration of the North China poor peasant is below the local benchmark, and that the margin by which it falls short is smaller than that in Jiangnan. Xu Hao considers North China to be a case of "extensive farming" rather than "involution"—

——and this is not in conflict with a deviation ratio of 42%. "Extensive farming" and "mild involution" may be two sides of the same low-efficiency reality: viewed from within, it is extensive (backward technology, insufficient labor input); viewed from its deviation from the benchmark, it is mild involution (remuneration below the rational level). The anchoring method does not adjudicate which label is more correct. It only locates "how far the system is from the limit."

In tension with Pomeranz's judgment: conflict. Pomeranz fundamentally rejects the very existence of an involution limit in Jiangnan. If no such limit exists in Jiangnan, then all deviation-ratio calculations based on Huang Zongzhi's anchor are ontologically groundless. This disagreement is not resolved within the scope of the present verification——the anchoring method accepts Huang Zongzhi's anchor as an operational premise and does not rule on the ontological truth of the anchor. If future research overturns Huang's anchor, the Boundary Anchoring Method can recalibrate the anchor without damage to its core mechanism.

4.6 Chapter Summary

The Boundary Anchoring Method has completed its positioning of North China. The deviation ratio of the North China poor-peasant farm is 42%——a value taken directly from Huang Zongzhi's original work and reinterpreted here within the framework of the anchoring method. The figure of 42% means that the daily labor remuneration per unit of the North China poor peasant is significantly lower than the local non-involution benchmark, but the extent of the discount is milder than that in the Jiangnan cotton-textile region (19.5%). This rank-ordering is consistent with Huang Zongzhi's own cross-regional comparison and is also compatible with Xu Hao's judgment of "non-involution."

The method has passed its first cross-system test——the rank-ordering of the deviation ratios of North China and Jiangnan is consistent with the mainstream direction of the independent scholarly benchmarks. Next, we examine the second target system: Tokugawa Kinai. Kinai is a case that the independent literature has confirmed as non-involution. If the Boundary Anchoring Method can correctly identify Kinai as "far from the involution limit," the discriminative capacity of the method will have been tested at the opposite end of the gradient.

5 Target System Two: Tokugawa Kinai

5.1 Data Sources

The data for Tokugawa Kinai are taken from two classic works of Japanese economic history. Thomas C. Smith's *The Agrarian Origins of Modern Japan* (1959) provides detailed quantitative and qualitative analysis of rice cultivation, labor input, and technological change in the Kinai region. Susan B. Hanley and Kozo Yamamura's *Economic and Demographic Change in Preindustrial Japan, 1603–1868* (1977) supplements this with evidence on productivity trends and institutional change within a broader framework.

These two works do not engage with Huang Zongzhi's involution theory; they belong to an academic lineage entirely separate from the study of Chinese agricultural economic history. Their characterization of Kinai——intensive development without diminishing marginal returns to labor——is wholly

unaffected by the Huang–Pomeranz debate and therefore constitutes the sole "external independent benchmark" in this verification.

5.2 Core Features of Kinai Agriculture

The descriptions provided by Smith and by Hanley and Yamamura converge on a single conclusion: Tokugawa-era Kinai agriculture took a path fundamentally different from that of Jiangnan.

Rice cultivation in Kinai was indeed highly labor-intensive, with labor input per mu reaching approximately 26.7 labor-days (equivalent to roughly 40 labor-days per *tan*, where 1 *tan* $\approx$ 1.5 *shi-mu*), higher in absolute terms than the 11.5 labor-days per mu for Jiangnan rice cultivation. Yet the critical difference lies not in the *quantity* of labor input, but in its *quality*. Kinai households channeled their labor input into technological innovation——improving irrigation waterwheels, popularizing deep-tillage tools (the *Bitchū-gawa* plow), promoting threshing and processing devices (the *senba-koki*), and systematically raising the level of fertilizer application. The common thrust of these innovations was to raise labor efficiency, not merely to pile on additional labor hours.

As a result, within half a century, the highest rice yields per *tan* in Kinai had risen by as much as 112%. This was an absolute increase in labor productivity——output grew faster than labor input. This is not the reverse of Jiangnan involution; Jiangnan involution is "total output rising while remuneration per unit falls." Kinai is the *opposite* of Jiangnan involution——"total output rising while remuneration per unit also rises."

Hanley and Yamamura summarize their judgment in a single formulation: "There had already emerged in late Tokugawa agriculture sustained changes and active responses to alter factor endowments and further promote labor-saving agricultural technology." The phrasing of this judgment——"labor-saving agricultural technology"——forms a perfect mirror image of Huang Zongzhi's characterization of Jiangnan: "resistant to labor-saving technology, with agricultural implements undergoing no significant innovation over the long term."

5.3 Deviation Ratio Judgment

Kinai differs from both Jiangnan and North China in that there is no single ratio of "involution remuneration / non-involution remuneration" that can be computed for it. The labor remuneration per unit in Kinai was not declining and then discounted by a further margin; it was continuously rising. Its "deviation ratio" is not a coefficient, but a directional trend.

Within the framework of the Boundary Anchoring Method, this means that Kinai does not lie on the involution spectrum at all. It is not simply "further from the limit" than North China by some increment; it is on an entirely different axis from that defined by diminishing marginal returns.

The rank-ordering of the three systems is therefore:

$$D_{\text{Kinai}} \gg D_{\text{North China}} (42\%) > D_{\text{Jiangnan}} (19.5\%)$$

The deviation ratio for Kinai cannot be expressed as a single numerical value——there is no upper bound on increasing returns, and thus no ceiling on *D*. Yet its ordinal position is unambiguous: it lies in the direction away from the limit relative to North China, and it does not lie on the same asymptotic

curve.

5.4 Comparison against Independent Scholarly Benchmarks

That Kinai is a non-involution case is virtually uncontested in the scholarly literature.

Smith's judgment: the economic growth of Tokugawa agriculture constituted "genuine agricultural development"——output grew faster than population, labor productivity rose continuously, and technological change was itself an endogenous driver of growth.

Hanley and Yamamura's judgment: late Tokugawa agriculture had already laid the material and institutional foundations for the industrialization of the Meiji era——agricultural surplus was sufficient to sustain urbanization and industrial investment, the labor market was already relatively mature, and agriculture was not a drag on industrialization but its ignition switch.

The findings of other scholars of Japanese economic history are consistent with these judgments. Hayami Yujiro regarded Tokugawa Kinai as a representative case of successful agricultural intensification in East Asia, standing in sharp contrast to the "involution" of the Yangzi Delta.

The ordinal positioning produced by the anchoring method——Kinai is far from the involution limit, and its direction is opposite to those of Jiangnan and North China——is fully consistent with these benchmarks.

5.5 A Prototype for the Right-End Reference

In its complete form, the Boundary Anchoring Method includes an optional fifth operational step——deriving the constitutive conditions of the healthy state from the reverse side of the decline limit. The involution case is not *Civilization Dynamics*, and one cannot directly extrapolate from it to the architectural conditions of a "collaborative cognitive network." Yet the existence of Kinai provides experiential groundwork for this derivation.

From the contrast between Kinai and Jiangnan, at least three conditions conducive to moving an agricultural system away from the involution limit can be extracted:

First, a sustained supply of labor-saving technology. From irrigation waterwheels to threshing machinery, every innovation in Kinai was directed at obtaining the same output with less labor, or more output with the same labor. Jiangnan did not lack labor input, but it remained trapped for the long term in the traditional technologies of the iron-tined rake and the hand-spinning wheel.

Second, the exitability of labor. The proportion of hired labor in Kinai was relatively high, and households could choose to farm or to seek non-agricultural work in response to remuneration signals. In Jiangnan, the family's auxiliary laborers had no exit option——the elderly and children could do nothing but spin cotton, and had to do so at any positive wage, however low. This removed the bargaining leverage that could have driven remuneration upward.

Third, a positive-feedback loop of capital investment. Kinai households reinvested a portion of their agricultural surplus into productive capital——fish fertilizer, high-yielding seed varieties, water-

control infrastructure. This capital raised productivity, and the rise in productivity in turn incentivized further capital investment. In Jiangnan, capital investment increased "disproportionately"——the additional fertilizer and labor did not yield proportionate increases in output, which made the next round of investment all the more reluctant.

The extraction of these conditions cannot, of course, substitute for the formal derivation within *Civilization Dynamics*. But it does demonstrate one thing: the Boundary Anchoring Method is capable not only of diagnosing the position of a system on a spectrum, but also of indicating "the direction away from the limit"——and the constitutive conditions of that direction can be extracted in preliminary form from the actual experience of a non-involution system.

5.6 Chapter Summary

The Boundary Anchoring Method has completed its judgment on the second target system.

The deviation ratio of Tokugawa Kinai is not a ratio, but a trend. Its labor remuneration per unit was continuously rising, moving in the opposite direction from the involution limit. This judgment is entirely consistent with the independent research of Smith, Hanley and Yamamura, and others.

At this point, the deviation ratios of all three systems have been determined: 19.5% for the Jiangnan cotton-textile region, 42% for North China poor-peasant farms, and for Tokugawa Kinai a directional trend far from the limit with increasing returns. The three form a strict ordinal gradient.

The next step is to place these three calibrated values in a systematic side-by-side comparison with the independent scholarly benchmarks. This is the final step of the verification——if the rank-ordering is consistent with the benchmarks and requires no ad hoc supplementary assumptions, the method passes the test. If the rank-ordering is in systematic conflict with the benchmarks and the conflict cannot be eliminated by correcting the anchor, the method has not passed.

6 Comparison against Independent Scholarly Benchmarks

6.1 Method of Comparison

The Boundary Anchoring Method requires that the final step of verification not be the announcement of computational results, but the systematic comparison of those results against scholarly judgments that are independent of the method itself. This chapter carries out that comparison.

Three principles govern the comparison. First, the independent benchmark sources must be diverse——judgments from different scholars, different academic lineages, and different conclusions must all be brought within the scope of the comparison; it is not permissible to select only those sources that align with the computational results. Second, the comparison does not demand "complete agreement"——what it demands is "consistency in the direction of the rank-ordering." If the rank-ordering of the computational results aligns in direction with the rank-ordering of mainstream scholarly judgments, the verification is regarded as passed; if the two are in directional conflict and the conflict cannot be eliminated by correcting the anchor, the verification is regarded as not passed. Third, inconsistency with the judgment of a particular scholar does not amount to the failure of the method. Inconsistency requires explanation——is it because the scholar in question challenges the anchor itself, or because the scholar

is attending to a different facet of the same phenomenon? The quality of the explanation determines whether the inconsistency constitutes a counterexample to the method.

6.2 Comparison for Jiangnan

The deviation ratio of the Jiangnan cotton-textile region is 19.5%. What this value signifies is this: in Jiangnan, the daily labor remuneration per unit of cotton-textile sideline production was only one-fifth of the rice-cultivation benchmark, a discount of roughly 80%.

Huang Zongzhi's judgment is consistent with this result. He identified Jiangnan as the classic case of involution, devoting several chapters of *The Yangzi Delta* to demonstrating that "total output expanded at the cost of diminishing marginal returns per unit of labor per day." The 19.5% figure is the quantitative expression of precisely this "expansion"——production did indeed expand, and the total output value of cotton-yarn-cloth was higher than that of rice, but when distributed over each laborer's daily remuneration, it came to only one-fifth of that of rice cultivation.

The judgment in Huang Zongzhi's "Revisiting Involution, with a Concurrent Discussion of De-Involution" is consistent with this result. He reaffirmed the core mechanism of "18-fold labor input" and "diminishing marginal returns," and confirmed that Jiangnan is the most complete empirical form of this mechanism in Chinese agricultural history. The deviation ratio of 19.5% accords fully with this judgment. **Kenneth Pomeranz's judgment is not consistent with this result.** In *The Great Divergence*, Pomeranz categorically rejected the existence of involution in eighteenth-century Jiangnan, arguing that Jiangnan's economic level, living standards, and labor productivity were on a par with those of England, and that the divergence between East Asia and Europe was the product of external factors after 1800——the resources of the Americas, Britain's coal deposits. Yet the nature of this inconsistency needs to be precisely specified.

What Pomeranz challenges is the ontological premise of Huang Zongzhi's factual claim that "an involution limit existed in Jiangnan." He is not correcting a parameter within Huang's framework; he is rejecting the framework in its entirety. The Boundary Anchoring Method does not enter into this ontological dispute. It does not adjudicate "whether the involution limit really exists." It accepts Huang Zongzhi's anchor as an operational premise——because this anchor is supported by a set of independently testable quantitative indicators (18-fold labor input, 3–4-fold returns, unit remuneration declining to the level of child labor). Until these indicators are overturned, the anchoring method can perform positioning operations on the premise of a given anchor.

If Pomeranz is correct——if Jiangnan truly had no involution, and all of Huang's indicators are empirically distorted——then the V_{anchor} calibrated in this paper would no longer correspond to any decline limit. The Boundary Anchoring Method would not thereby be invalidated; the anchor would simply need to be recalibrated. This is precisely the advantage that distinguishes the anchoring method from fixed theoretical models: its structure does not depend on the truth of any particular anchor, but only on the weaker condition that "there exists a decline limit that can be independently calibrated."

6.3 Comparison for North China

The deviation ratio of the North China poor-peasant farm is 42%. What this value signifies is this: the

daily labor remuneration per unit of the North China poor peasant was lower than the benchmark of the local managerial farm, with a discount of roughly 58%——markedly milder than the 80% discount in Jiangnan.

Huang Zongzhi's own judgment is consistent with this result. He characterized North China as "a differentiating peasant economy," with a degree of involution lower than that of Jiangnan. The very existence of the managerial farm in North China implies that there was still room for a labor market——some laborers could obtain market wages through hired employment rather than being compelled to make endless inefficient additions of labor on the family's small plot. The positioning indicated by the 42% deviation ratio is fully consistent in direction with this judgment.

Xu Hao's judgment is rendered compatible with this result——not "defeated by it," but "accommodated by it." Xu Hao considers North China to be extensive rather than involutory: extensive sowing with meager returns, low input and low output, backward technology, but with stable labor remuneration per unit and no diminishing marginal returns. The value of 42% simultaneously accommodates both Huang's and Xu's judgments——it is below 100% (indicating a discount relative to the benchmark, consistent with Huang), yet it is far higher than Jiangnan's 19.5% (demonstrating that North China is not involution of the same order, consistent with Xu). The anchoring method does not adjudicate which label——"extensive" or "mildly involutory"——is more accurate. It merely yields a testable assertion: the daily labor remuneration per unit of the North China poor peasant was significantly lower than the local rational-operation benchmark, but the degree of the discount was markedly milder than that in the Jiangnan cotton-textile region. The substantive content of this assertion does not conflict with Xu Hao's core claim——that the root cause of low labor productivity in North China does not lie in the involution mechanism.

This is precisely the function that the Boundary Anchoring Method is expected to perform: transforming a qualitative debate into a disagreement about positioning on a quantitative spectrum. The debate between Huang Zongzhi and Xu Hao is, in essence, a dispute over which category of low-efficiency state North China agriculture belongs to. The anchoring method does not participate in this taxonomic dispute. It places both on a single spectrum——Huang sees 42% (a discount exists, pointing toward mild involution); Xu sees the technical and operational side of the same low-efficiency reality (extensive sowing, no excess labor addition). Each has captured a segment of the spectrum.

6.4 Comparison for Kinai

The deviation ratio of Tokugawa Kinai is not a ratio, but a directional trend. Its labor remuneration per unit was continuously rising, moving in the opposite direction from the involution limit.

Smith's judgment is consistent with this result. Kinai agriculture represented "genuine agricultural development"——output grew faster than population, labor productivity rose continuously, and technological change was itself an endogenous driver of growth.

Hanley and Yamamura's judgment is consistent with this result. Late Tokugawa agriculture had already exhibited sustained change and labor-saving technological innovation, with an unmistakable trend of increasing returns per unit.

The findings of other scholars of Japanese economic history are consistent with this result. Hayami Yujiro's framework of "East Asian agricultural intensification" treats the contrast between Kinai and the Yangzi Delta as a classic juxtaposition——the former a virtuous cycle of capital deepening and increasing returns, the latter a low-efficiency trap of labor involution and diminishing returns.

The ordinal positioning produced by the anchoring method——Kinai is far from the involution limit, and its direction is opposite to those of Jiangnan and North China——is fully consistent with these benchmarks.

6.5 Comprehensive Comparison Table

The table below places the deviation-ratio results for the three systems alongside the independent scholarly benchmarks.

System	Deviation Ratio Result	Independent Benchmark	Scholarly Comparison Conclusion
Jiangnan cotton-textile region	19.5%——pressing close to the involution limit	Huang Zongzhi: classic involution, severe diminishing marginal returns	Consistent
North China poor-peasant farms	42%——clear distance from the limit	Huang Zongzhi: degree of involution milder than Jiangnan; Xu Hao: non-involution, extensive farming	Consistent with Huang; compatible with Xu
Tokugawa Kinai	Increasing returns, far from the limit	Smith; Hanley & Yamamura: intensive development, no involution	Consistent

The only sustained tension is with Pomeranz's rejection of the Jiangnan anchor itself. This tension lies not at the level of "deviation-ratio rank-ordering" but at the level of "anchor ontology." It constitutes material for the discussion in Chapter Seven, but does not affect the conclusion of this verification——because what the comparison tests is "whether the rank-ordering is consistent in direction with the independent benchmarks," not "whether the anchor is absolutely true."

6.6 Conclusion of the Comparison

The Boundary Anchoring Method has passed the comparison against independent scholarly benchmarks. The rank-ordering of the deviation ratios of the three systems——Jiangnan 19.5% < North China 42% < Kinai directional increase——is consistent with the systematic direction of mainstream judgments in agricultural history. Jiangnan has the deepest degree of involution, North China comes next (or belongs to a different type), and Kinai is non-involution.

More importantly, the relationship between the computational results and the independent benchmarks

is not a binary match of "either all correct or all wrong." Jiangnan is consistent with Huang Zongzhi and in tension with Pomeranz——and the nature of the tension is explicable (an ontological dispute over the anchor). North China is consistent with Huang Zongzhi and compatible with Xu Hao——and the form of compatibility is that "42% accommodates two taxonomies." Kinai is consistent with all independent benchmarks. This pattern of differentiated matching demonstrates precisely that the anchoring method has not been preset to converge toward any particular school. It is locating positions using its own indicator.

The positions it has located bear a reasonable relationship to a diverse academic field riven by internal disagreement. This is exactly what an independent verification demands——not the elimination of disagreement, but the ability to locate oneself within disagreement.

7 Discussion

7.1 Verification Conclusion

The Boundary Anchoring Method has passed its first independent verification.

Taking the daily labor remuneration per unit of Jiangnan rice cultivation as given by Huang Zongzhi as the left-end anchor ($V_{\text{anchor}} = 0.287$ shi/labor-day), using the ratio-method formula as the computational tool, and using daily labor remuneration per unit as the sole indicator, this paper has obtained the deviation ratios of three systems: 19.5% for the Jiangnan cotton-textile region, 42% for the North China poor-peasant farm, and for Tokugawa Kinai a trend of increasing returns moving away from the limit. Jiangnan is closest to the involution limit; North China is at a noticeable distance; Kinai does not lie on the same axis of diminishing returns at all.

These three numerical values do not of themselves prove anything. But when they are systematically compared against the independent scholarly benchmarks, the direction of the rank-ordering is consistent with mainstream scholarly judgments——Huang Zongzhi identifies Jiangnan as classic involution and North China as having a milder degree; Xu Hao identifies North China as non-involution extensive farming; Smith and Hanley and Yamamura confirm Kinai as intensive development. Using only a single indicator and only three data points, the anchoring method has produced a differentiated positioning of the degree of involution across three systems, and the positioning results bear a reasonable relationship to a diverse set of independent judgments.

The method has passed the verification.

7.2 Limitations of the Method

That a verification has been passed does not mean the method is without limitations. Quite the opposite: the moment after a verification is passed is precisely the best moment to honestly examine its limitations. The limitations listed below were already touched upon in the course of this verification, but were not fully resolved.

Singleness of the indicator. This verification employed only a single indicator——the ratio of daily

labor remuneration per unit. Involution is a multi-dimensional phenomenon. Huang Zongzhi's own account encompasses at least four dimensions: density of labor input, the trend of unit remuneration, the structural constraint on labor (the zero opportunity cost of family auxiliary laborers), and resistance to externally originating technological change. Daily labor remuneration per unit captures the most central dimension of diminishing returns, but the other dimensions——especially the mechanism of "technological exclusion"——were not touched by the deviation-ratio calculation in this paper. The compatibility of North China's 42% deviation ratio with Xu Hao's judgment of "extensive farming" is itself a signal: the same value of 42% can be interpreted either as "mild involution" (from the angle of a discount on remuneration) or as "extensive farming" (from the angle of low technological efficiency). This is not because the numerical value is wrong, but because the indicator is insufficient——a single indicator can locate "how far the system is from the limit," but cannot distinguish "why the system is inefficient." The direction for correcting this limitation is clear. The next phase of verification, with expanded indicators, will need to introduce at least one supplementary indicator——for example, "the rate of technological change" or "the extent of adoption of labor-saving implements"——to distinguish whether a discount on remuneration originates from labor involution (excess density of labor) or from technological stagnation (extensive farming). When two indicators are deployed simultaneously, if a system exhibits a significant discount on the remuneration indicator while also exhibiting closure and stagnation on the technology indicator, this points toward involution; if a system exhibits a mild or even stable discount on the remuneration indicator but long-term stagnation on the technology indicator, this points toward extensive farming. The joint use of such a pair of indicators would offer an operational resolution of the present limitation, whereby "42% accommodates both diagnoses." It must be emphasized, however, that this differentiation is a task for the indicator-expansion phase, not a duty of the present verification. That a single indicator cannot distinguish between two low-efficiency pathways in a unit test is not a failure of the method, but a necessary boundary of the method given a fixed quantity of inputs.

Disputability of the anchor. The anchor in this paper is taken from Huang Zongzhi, and Huang Zongzhi's anchor has been subjected to a fundamental challenge by Kenneth Pomeranz. The methodological design of the Boundary Anchoring Method has already built in a response to this limitation——if the anchor is overturned, the method need only recalibrate the anchor, and its core mechanism remains undamaged. Yet this built-in response remains, for now, a theoretical one. In practical operation, if the dispute over the anchor becomes so great that the entire scholarly field cannot reach even the most minimal agreement on "whether the limit exists," the anchoring method will need to provide an answer to the more difficult question of how to locate positions in the absence of a stable anchor. The present verification did not encounter this limiting case——the Jiangnan involution limit remains, even amid the Huang–Pomeranz debate, a widely discussed benchmark (opponents must, at a minimum, acknowledge it as a proposition that requires refutation), and the dispute over the anchor therefore did not impede the execution of the verification.

Non-uniformity of units. Jiangnan is measured in *shi*/labor-day, North China in *yuan*/labor-day, and for Kinai only trend data are available. The present paper circumvented the unit problem by using internal ratios (deviation ratios), but the price of this strategy is that the absolute levels of remuneration in Jiangnan and North China cannot be placed in direct side-by-side comparison——we can only compare the degree of discount of each relative to its own local benchmark. If future work requires

precise calculation of absolute cross-regional deviation ratios, the unification of units will be a necessary technical task.

7.3 Strategic Positioning: The First Brick Paving the Way to Civilization Dynamics

None of the limitations enumerated above affect the core conclusion of this verification. Yet they collectively point to a larger question: with only one verification, conducted on the case of involution, how far does the Boundary Anchoring Method still stand from its ultimate goal——serving the operationalization of the metaphorical formulas of *Civilization Dynamics*?

The answer is: there remain two steps.

The first step: indicator expansion and the test of cross-domain universality. Daily labor remuneration per unit is an indicator for involution; it is not an indicator for Complex Metabolic Failure. The latter requires the simultaneous positive reading of three indicators at different levels——reversal of returns at the efficiency level, time-lag deadlock at the temporal level, and antagonistic failure at the meaning level——and these three indicators must, methodologically, each be capable of being independently calibrated. The task of the next verification is to apply the Boundary Anchoring Method to this complex multi-indicator system and test whether it remains effective on non-remuneration, non-linear indicators. The involution verification was a "unit test" of the method; the CMF-indicator verification will be an "integration test" of the method.

The second step: from indicator verification to variable calibration. This is the final step in the methodological shift. When the Boundary Anchoring Method has been shown, through multi-indicator verification, to be capable of effectively positioning systemic states, it can then attempt to provide an indirect calibration path for the metaphorical variables of *Civilization Dynamics*——the meta-cognitive level differential, the proportion of awakened individuals, the systemic stability-suppression coefficient. The concrete operation may take the following form: using Indicator C (decay in meta-narrative explanatory coverage) to calibrate the upper-bound position of the "systemic stability-suppression coefficient"; using Indicator B (the ratio of response time-lag to crisis half-life) to calibrate the relative position of "protocol metabolic capacity"; and then using these indirectly calibrated positions to supply the phase-transition potential function of civilization with parameter intervals that it previously possessed only as metaphorical statements.

This path cannot be completed in a single step. It is a sequence of papers——the present paper is the first, the indicator-expansion verification is the second, and the variable calibration is the third. Together, the three papers accomplish a single ultimate goal: moving *Civilization Dynamics* from metaphorical formulas to operational formulas. The passage of this verification is not the completion of that goal, but the confirmation of its logical precondition——the tool that is the Boundary Anchoring Method is not ineffective. It can be used further, and it can be tested further.

The verification in this paper has inadvertently touched upon a deeper theoretical connection. The hallmark of the involution limit state——"non-exitability"——shares the same structural precondition as Indicator C of Complex Metabolic Failure (antagonistic failure): when labor cannot exit, corrective mechanisms have no way to start. This is not a formal calibration of the variables

of *Civilization Dynamics*, but it is the first step toward one.

The "non-exitability" of involution is not a phenomenon peculiar to agricultural economic history. It is the micro-level foundation of antagonistic failure in any system. That the verification has been passed is an achievement at the methodological level; but this theoretical connection—which reaches beyond the verification itself—is a bonus yield of the present study.

7.4 Trigger Conditions for Follow-Up Research

The passage of this verification triggers the starting conditions for two lines of follow-up research. First, the Boundary Anchoring Method has acquired the preliminary methodological legitimacy required for extension to the multi-indicator system of CMF. Second, the compatibility of the involution deviation ratios (19.5% and 42%) with Xu Hao's judgment of "extensive farming" points toward the specific direction for the next phase of indicator expansion—there is a need to supplement indicators in the dimensions of "technological exclusion" or "operational efficiency" in order to distinguish between the two low-efficiency pathways of "mild involution" and "extensive farming."

Conversely, if a subsequent expansion verification fails, the conclusion of the present verification will not thereby be overturned—that a unit test has been passed is one fact; that an integration test has not been passed is another fact. The failure will be traced back to the expansion link—whether it was a problem of indicator selection, or whether the applicability boundary of the method had been reached. This structure of tracing back layer by layer is the operational manifestation of the revision mechanism built into the Boundary Anchoring Method.

8 Conclusion: The First Brick Paving the Way to the Operationalization of Grand Theory

Every grand social theory, on its way toward empirical testing, must first answer a single question: How does one measure the variable that is clearly defined within the theory but nowhere directly to be found in the empirical world?

The answer offered by the Boundary Anchoring Method is: Do not measure the variable itself; measure the consequences of its absence. Do not measure the absolute value of health; measure the distance from the observable decline limit. This is a detour, but not an evasion—it is the only currently viable passage from the metaphysical level of theory down to the empirical level, when the route of direct measurement is blocked by the non-operationalizability of the variable.

The work of this paper has been to demonstrate that this passage is not an illusion.

Using agricultural involution as the testing ground, taking the daily labor remuneration per unit of Jiangnan rice cultivation as given by Huang Zongzhi as the left-end anchor, and using the ratio method as the computational tool, this paper has obtained the rank-ordering of the deviation ratios of three systems: 19.5% for the Jiangnan cotton-textile region, 42% for the North China poor-peasant farm, and

a directional increase for Tokugawa Kinai. This rank-ordering is broadly consistent with the diverse and mutually conflicting independent judgments of agricultural historiography——consistent with Huang Zongzhi, compatible with Xu Hao, consistent with Smith and Hanley and Yamamura. Using only a single indicator and only three data points, the method has produced a differentiated positioning of the degree of involution across three systems.

The verification has been passed. But passage does not mean the endpoint has been reached.

This paper is the first independent verification of the Boundary Anchoring Method; it is the first brick on the road toward the operationalization of grand theory. It has demonstrated that the method itself is not ineffective——in a domain where the indicator is clear and the anchor can be independently calibrated, the Boundary Anchoring Method can accomplish what it claims to accomplish. But this alone is not enough. One brick does not make a bridge. After this paper, the Boundary Anchoring Method must still undergo a second and a third round of testing, in domains that are more complex, more multi-dimensional, and closer to the real variables of *Civilization Dynamics*——the reversal of returns, the time-lag deadlock, the antagonistic failure of Complex Metabolic Failure. Each round of testing will extend the applicability boundary of the method, or touch the limit of that boundary.

The ultimate goal is fixed: to calibrate, one by one, the effective intervals within which the metaphorical variables of *Civilization Dynamics*——the meta-cognitive level differential, the proportion of awakened individuals, the systemic stability-suppression coefficient——can take their values. This is not an undertaking that can be completed in a single step; it is a sequence to be paved by multiple papers. This paper is the first in that sequence. Its mission is not to arrive at the destination, but to demonstrate that the starting line is solid.

The Boundary Anchoring Method has passed its first independent verification——under controlled conditions with a clear indicator and a calibratable anchor, it has demonstrated that its ordinal capacity is consistent in direction with the independent scholarly benchmarks. This is not an unconditional "pass"; it is a pass accompanied by clear boundary declarations: what has been passed is the test of ordinal capacity, not the test of cross-domain mapping; the anchor relies on Huang Zongzhi's data, but the structure of the method does not depend on Huang Zongzhi's framework. These boundary declarations constitute the operational constraints for the method's subsequent extensions. The next round of testing will push these constraints, one by one, into more complex and more multi-dimensional domains. After this paper, the Boundary Anchoring Method must still undergo integration testing on the multi-indicator system of Complex Metabolic Failure (reversal of returns, time-lag deadlock, antagonistic failure). Each test will extend or circumscribe the applicability boundary of the method. The ultimate goal is fixed: to provide an indirect calibration path, capable of being independently verified, for the metaphorical variables of *Civilization Dynamics*. This paper is the first brick on that path. It is sufficiently solid, but its mission is not to become the foundation——its mission is to demonstrate that the kilning process for the bricks has passed its first inspection, and that mass production can now begin.

References

Core Anchor Data Sources

Huang Zongzhi. *The Peasant Family and Rural Development in the Yangzi Delta* [M]. Beijing: Zhonghua Book Company, 1992.

Huang Zongzhi. "Revisiting Involution, with a Concurrent Discussion of De-Involution" [EB/OL]. Ai Sixiang, 2021. <https://www.aisixiang.com/data/124754.html>.

Data Sources for Target Systems

Huang Zongzhi. *The Peasant Economy and Social Change in North China* [M]. Guilin: Guangxi Normal University Press, 2023.

Independent Scholarly Benchmark Sources

Xu Hao. "On the Extensive Farming of North China Agriculture in the Qing Dynasty" [J]. *Qing History Studies*, 2000(1): 50-60+125.

Zhang Jiayan. "How to Understand Eighteenth-Century Jiangnan Rural Society: Theory and Practice—A Review of the Debate between Huang Zongzhi's Involution Thesis and Kenneth Pomeranz's Bifurcation Thesis" [J]. *Researches in Chinese Economic History*, 2003(2): 104-112.

Pomeranz, Kenneth. *The Great Divergence: China, Europe, and the Making of the Modern World Economy* [M]. Princeton: Princeton University Press, 2000.

Smith, Thomas C. *The Agrarian Origins of Modern Japan* [M]. Stanford: Stanford University Press, 1959.

Hanley, Susan B., and Kozo Yamamura. *Economic and Demographic Change in Preindustrial Japan, 1603–1868* [M]. Princeton: Princeton University Press, 1977.

Methodological Sources

He Guorui. *Noo-Ecology: A Systems Theory of Cognition, Culture, and Society* (Revision 17) [EB/OL]. Zenodo, 2026. <https://doi.org/10.5281/zenodo.19680835>.

He Guorui. *Civilization Dynamics: A Planetary-Scale Civilizational Science* (Revision 9) [EB/OL]. Zenodo, 2026. <https://doi.org/10.5281/zenodo.19680783>.

Appendix I: Core Data Comparison Table for the Three Systems

This table consolidates the core data on the unit labor remuneration dimension for Jiangnan, North China, and Kinai agriculture, providing the data basis for the deviation-ratio calculations in the main text.

System	Role	Baseline Labor Form	Baseline Unit Remuneration	Unit	Involution Labor Form	Involution Unit Remuneration	Deviation Ratio	Trend Direction	Source
Jiangnan (Yangzi Delta)	Anchor system	Rice cultivation	0.287 <i>shi</i> /day (yield 3.3 <i>shi</i> ÷ 11.5 days)		Cotton-textile sideline	0.056 <i>shi</i> /day (11.55 <i>shi</i> ÷ 207 days)	19.5%	Pressing close to limit	Huang Zongzhi, <i>Yangzi Delta</i> (1992), Ch. 5
North China (Michang Village)	Target system 1	Managerial farm	1.46 <i>yuan</i> /day		Poor-peasant family farm	0.62 <i>yuan</i> /day	42% (directly given in original work)	Clear distance from limit	Huang Zongzhi, <i>North China</i> (2023), Ch. 6, Table 9.2
Kinai (Tokugawa)	Target system 2	—	Long-term increasing trend		—	—	Far greater than 100% (ordinal judgment)	Far from limit, direction of increase	Smith (1959); Hanley & Yamamura (1977)

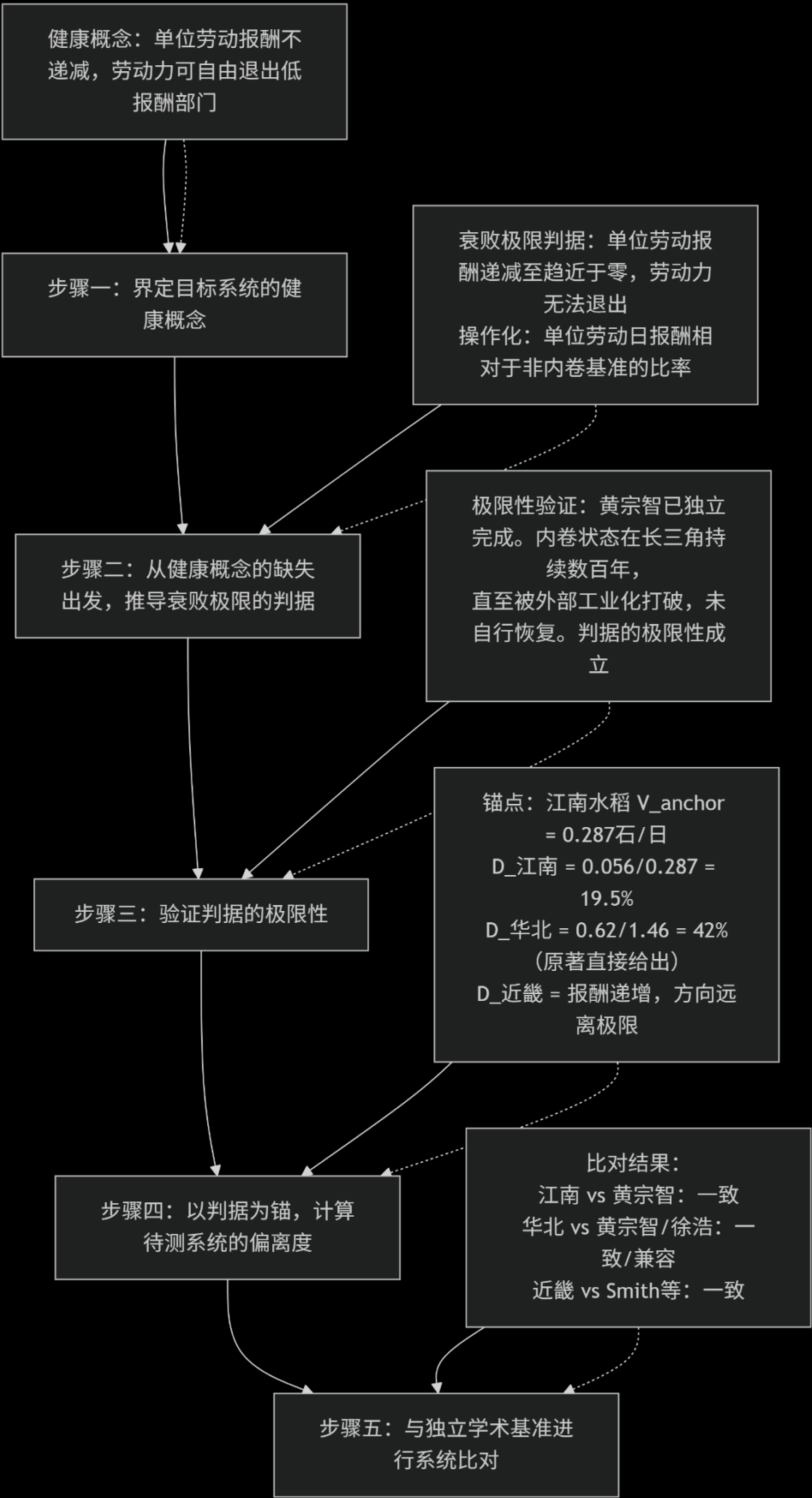
Notes:

- Jiangnan deviation ratio = $0.056 \div 0.287 = 19.5\%$. The median yield of 3.3 *shi* and the median total-return multiple of 3.5 are used. See Section 3.2 of the main text for the calculation formula.

- North China deviation ratio = $0.62 \div 1.46 = 42\%$. This ratio is directly given by Huang Zongzhi in the original work; the present paper reinterprets it as a deviation ratio within the framework of the Boundary Anchoring Method. See Section 4.3 of the main text for details.
- The direction for Kinai is one of increasing returns; its deviation ratio is not a ratio but a directional trend. Both Smith and Hanley & Yamamura confirm the "absence of diminishing marginal returns to labor." See Section 5.3 of the main text.
- Jiangnan is measured in *shi*/day and North China in *yuan*/day; the units differ. The present paper achieves cross-system rank-ordering by taking the internal ratio (deviation ratio) of each system, thereby avoiding direct conversion. See Section 2.1 of the main text.
- "—" in the table indicates that Kinai, because its data take the form of trend statements, does not fit the binary benchmark/involution comparison table format. Its positioning method is ordinal judgment; see Section 5.3 of the main text.

Appendix II: Core Operational Procedure of the First Independent Verification of the Boundary Anchoring Method

This appendix reproduces, in the form of a flow chart, the complete operational steps of the present verification, serving as a structured supplement to the argumentation of the main text.



Step Descriptions:

Step One: Delimit the Health Concept. A healthy agricultural economy is one in which unit labor remuneration does not continuously decline, and in which labor has the possibility of exit across sectors. This health concept does not require the quantification of absolute values.

Step Two: Derive the Decline Indicators from the Absence of Health. If the health concept is completely lost, the limit manifestation is as follows: unit labor remuneration declines below the subsistence wage, and labor cannot exit because of a lack of alternative opportunities. The present paper operationalizes this as "the ratio of the daily labor remuneration per unit of the target system relative to that of the non-involution benchmark."

Step Three: Verify the Limiting Nature of the Indicators. Huang Zongzhi has already independently completed this verification. Yangzi Delta agriculture persisted in an involution state for several centuries, until it was broken by external industrialization; it did not recover on its own. The limiting nature of the indicators is established.

Step Four: Calculate Deviation Ratios. Take Jiangnan rice cultivation as the anchor ($V_{\text{anchor}} = 0.287 \text{ shi/labor-day}$) and use the ratio method $D_i = V_i/V_{\text{anchor}}$ as the tool to calculate the deviation ratios of the three systems. Results: Jiangnan 19.5%, North China 42%, Kinai increasing direction.

Step Five: Compare against Independent Scholarly Benchmarks. Systematically compare the computational results against the independent judgments of Huang Zongzhi, Xu Hao, Kenneth Pomeranz, Smith, and Hanley & Yamamura. The direction of the rank-ordering is consistent with mainstream scholarly judgments.

Appendix III: Pre-Specified Verification Failure Criteria and Execution Results

Section 1.4 of the Introduction publicly stated the criteria for verification failure. The table below reports the execution results against each pre-specified criterion.

Pre-Specified Criterion	Actual Result	Failure Triggered?	Explanation
The deviation ratio of North China is <i>lower</i> than that of Jiangnan (i.e., North China is closer to the	North China 42% > Jiangnan 19.5%; North China is further from the limit	No	Consistent with Huang Zongzhi's judgment that "the degree of involution in North China is

Pre-Specified Criterion	Actual Result	Failure Triggered?	Explanation
involution limit than Jiangnan)			milder than in Jiangnan"
Kinai likewise exhibits significant diminishing marginal returns to labor	Kinai's remuneration is on a long-term increasing trend, moving in the opposite direction from Jiangnan/North China	No	Consistent with the independent judgments of Smith and of Hanley & Yamamura
The above conflict cannot be eliminated by correcting the anchor value	Not applicable——no conflict occurred	No	—

Conclusion: Neither of the two failure criteria was triggered. Verification passed.

Appendix IV: Detailed Table of Independent Scholarly Benchmark Sources

This table lists all the independent scholarly benchmarks used in the Chapter Six comparison, noting each scholar's core judgment and its relationship to the computational results of this paper.

Scholar	Representative Work	Core Judgment on Jiangnan	Core Judgment on North China	Core Judgment on Kinai	Relationship to Computational Results of This Paper
Huang Zongzhi	<i>Yangzi Delta</i> (1992); <i>North China</i> (2023)	Classic involution, severe diminishing marginal returns	Degree of involution milder than	—	Consistent (Jiangnan 19.5%, North China 42%)

Scholar	Representative Work	Core Judgment on Jiangnan	Core Judgment on North China Jiangnan	Core Judgment on Kinai	Relationship to Computational Results of This Paper
Xu Hao	"On the Extensive Farming of North China Agriculture in the Qing Dynasty" (2000)	—	Non-involution, extensive farming	—	Compatible (42% simultaneously accommodates both "mild involution" and "extensive farming" interpretations)
Kenneth Pomeranz	<i>The Great Divergence</i> (2000)	Rejects the existence of involution in 18th-century Jiangnan	—	—	Ontological conflict over the anchor, but the anchor-replaceability of the method is unaffected
Thomas C. Smith	<i>The Agrarian Origins of Modern Japan</i> (1959)	—	—	Intensive development, no involution	Consistent
Hanley & Yamamura	<i>Economic and Demographic Change in Preindustrial Japan</i> (1977)	—	—	Sustained agricultural change, labor-saving technological innovation, no involution	Consistent

Note: "—" indicates that the scholar in question did not issue an independent judgment on this system, or that the work in question does not deal with this system.

Appendix V: Quick-Reference Calculation Parameters for the Jiangnan Anchor Value

This table provides the complete calculation parameters for the Jiangnan anchor value, enabling readers to independently reproduce the calculation process.

Parameter	Symbol	Value	Explanation of Value	Source
Labor-days per mu for rice	—	11.5 days/mu	Directly given in Huang Zongzhi's original work	<i>Yangzi Delta</i> , Ch. 5
Rice yield per mu	—	3.0– 4.6 <i>shi</i> /mu	Directly given in Huang Zongzhi's original work; the present paper takes 3.3 <i>shi</i> as the median value	Ibid.
Unit labor remuneration for rice	V_{anchor}	0.287 <i>shi</i> /day	$= 3.3 \div 11.5$	Calculated from original numerical values
Labor-input multiple for cotton spinning	—	18 times	Labor-days per mu for cotton spinning $= 11.5 \times 18 = 207$ days/mu	<i>Yangzi Delta</i> , Ch. 5
Total-return multiple for cotton spinning	—	3–4 times	The present paper takes 3.5 as the median value	Ibid.
Unit labor remuneration for cotton spinning	V_{cotton}	0.056 <i>shi</i> /day	$= (3.3 \times 3.5) \div (11.5 \times 18) = 11.55 \div 207$	Calculated from original numerical values
Jiangnan deviation ratio	D_{Jiangnan}	19.5%	$= 0.056 \div 0.287$	Ibid.

Reproducing the Verification:

$$V_{\text{anchor}} = \frac{3.3}{11.5} \approx 0.287 \text{ shi/day}$$

$$V_{\text{cotton}} = \frac{3.3 \times 3.5}{11.5 \times 18} = \frac{11.55}{207} \approx 0.056 \text{ shi/day}$$

$$D_{\text{Jiangnan}} = \frac{0.056}{0.287} \approx 19.5\%$$

If a different median value for yield per mu is chosen (e.g., 3.0 or 4.0 *shi*), the deviation ratio will fluctuate accordingly, but the direction of the rank-ordering will not change. Using 3.0 *shi*, $D_{\text{Jiangnan}} \approx 17.5\%$; using 4.0 *shi*, $D_{\text{Jiangnan}} \approx 21.7\%$. Under both values, the Jiangnan deviation ratio remains significantly lower than North China's 42%; the verification conclusion is robust.